

1 The bacterium *Escherichia coli* is a typical prokaryotic cell.

- (a) One key feature of a prokaryotic cell is the absence of organelles surrounded by a double membrane.

Name **one** organelle surrounded by a double membrane that is **not** found in *E. coli*.

..... [1]

- (b) Bacteriophages are viruses that infect bacterial cells. They are specific to a bacterial species or to a strain of a bacterium.

Figure 1.1 is a transmission electron micrograph showing a type of bacteriophage, known as a T-phage, attached to the surface of *E. coli*. T-phages are also visible inside the cell.

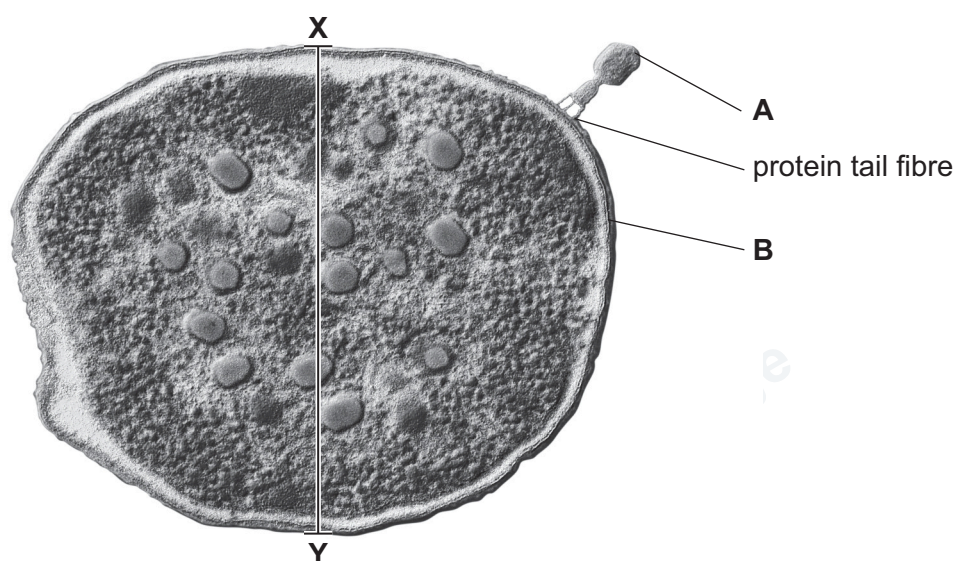


Figure 1.1

Structure **A** of the T-phage in Figure 1.1 contains the nucleic acid core, which is DNA. The DNA is injected into *E. coli* after the T-phage has attached to structure **B**, the outer layer of the cell.

- (i) Name structure **A** and structure **B**, shown on Figure 1.1.

A

B

[2]

- (ii) The actual diameter of the *E. coli* cell in Figure 1.1 along the line **X–Y** is $0.9\mu\text{m}$.

Calculate the actual length of the T-phage attached to the surface of *E. coli*.

Give your answer in nanometres (nm) to the nearest **whole nm**.

actual length = nm [2]

- (c) Suggest why a T-phage can attach to *E. coli*, but cannot attach to cells of other species of bacteria.

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..... [2]

- (d) Research has been carried out into the use of bacteriophages as a treatment for infectious diseases (bacteriophage therapy), such as those caused by some strains of *E. coli*.

It has been suggested that bacteriophage therapy may help to solve some of the problems associated with the current treatments for bacterial diseases.

- (i) Cholera, malaria and tuberculosis (TB) are three common infectious diseases.

Explain which of these three diseases is **least** likely to be treated with bacteriophage therapy.

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..... [1]

- (ii) Suggest the benefits of using bacteriophage therapy for bacterial diseases.

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..... [3]

[Total: 11]